

Every Algebraic Number is a Critical Value of an Integer Polynomial

by RJ Acuña

Abstract

For every nonconstant $f \in \mathbb{Z}[X]$ we construct $g \in \mathbb{Z}[X]$ and an irreducible $H \in \mathbb{Z}[X]$ with $H \mid g'$ and $H^2 \mid f \circ g$. Equivalently, every algebraic number is a critical value of an integer polynomial, so the set of critical values of $\mathbb{Z}[X]$ is exactly $\overline{\mathbb{Q}}$. The construction is explicit. The proof was assisted by AI, and has been formalized in Lean.

1 Introduction

A critical value of a nonconstant $g \in \mathbb{Z}[X]$ is a number α with $g(\beta) = \alpha$ and $g'(\beta) = 0$ for some β . Since $\overline{\mathbb{Q}}$ is closed under polynomial evaluation, every critical value is algebraic. This note proves the converse, and does so with an explicit construction.

Theorem 1. *Let $f \in \mathbb{Z}[X]$ be nonconstant. Then there exist $g, H \in \mathbb{Z}[X]$ with H irreducible in $\mathbb{Z}[X]$ and $1 \leq \deg H$, such that*

$$H \mid g' \quad \text{and} \quad H^2 \mid f \circ g \quad \text{in } \mathbb{Z}[X].$$

The g produced satisfies $\deg g \geq 2$.

The corollary is the statement in the title.

Corollary 1. *The set of critical values is exactly $\overline{\mathbb{Q}}$.*

1.1 Conventions

Throughout, p' = derivative p , and for a fixed nonzero $M \in \mathbb{Z}$ we write

$$\tilde{p} := p(MX) = p \circ (MX),$$

the *dilation* of p . Dilation is a ring homomorphism $\mathbb{Z}[X] \rightarrow \mathbb{Z}[X]$, it is injective for $M \neq 0$, it preserves degree, and it interacts with the derivative by the chain rule

$$(\tilde{p})' = M \tilde{p}'. \tag{1}$$

These four facts are used on essentially every line of §3 and are worth isolating before anything else; the factor M in (1) is the reason M has to appear in the statement of Lemma 2.

2 The construction

Fix a nonconstant $f \in \mathbb{Z}[X]$ and suppose $X \nmid f$; the excluded case is settled separately in §6. All of the following data is explicit and computable.

<i>Symbol</i>	<i>Definition</i>	<i>Notes</i>
h	a primitive irreducible factor of f with $h(0) \neq 0$	in $\mathbb{Z}[X]$
f_1	the cofactor, $f = hf_1$	in $\mathbb{Z}[X]$
n	$\deg h \geq 1$	
h_0	$h(0) \neq 0$	in $\mathbb{Z}$
u, v, ρ	any Bézout data, $uh + vh' = \rho, \rho \neq 0$	$u, v \in \mathbb{Z}[X], \rho \in \mathbb{Z}$
M	ρh_0 , so $M \neq 0$	the dilation factor
H	$h(MX)/h_0$	$H(0) = 1, \deg H = n$
W	$(v(0)H - v(MX))/\rho$	$W(0) = 0$
g	$MX + h(MX)W = MX + h_0HW$	

Existence of h . f is nonconstant, so it has an irreducible factor. Irreducible factors of positive degree in $\mathbb{Z}[X]$ are primitive, and $h(0) \neq 0$ because $h(0) = 0$ would give $X \mid h \mid f$, excluded.

Existence of the Bézout data. h is irreducible and primitive in $\mathbb{Z}[X]$, hence irreducible in $\mathbb{Q}[X]$ by Gauss's lemma. In characteristic zero $h' \neq 0$ and $\deg h' < \deg h$, so $h \nmid h'$ and therefore $\gcd(h, h') = 1$ in the principal ideal domain $\mathbb{Q}[X]$. Bézout there and clearing denominators produces $u, v \in \mathbb{Z}[X]$ and $\rho \in \mathbb{Z} \setminus \{0\}$.

Any nonzero ρ in the elimination ideal $(h, h') \cap \mathbb{Z}$ works. The resultant $\text{Res}(h, h')$ is one admissible choice and is never needed as such; smaller choices give a smaller g , and the generator of the elimination ideal — the *reduced* resultant — is the best available. Example 2 below uses $\rho = 2$ where the resultant is 4.

2.1 Integrality

The two divisions in the table are exact, which is the whole reason M was defined as the product ρh_0 rather than as either factor alone.

Lemma 1 (Integrality). $H \in \mathbb{Z}[X]$ with $H(0) = 1$ and $\deg H = n$; and $W \in \mathbb{Z}[X]$ with $W(0) = 0$.

Proof. Write $h = \sum_j h_j X^j$ and $v = \sum_j v_j X^j$. The coefficients of H are $H_0 = h_0/h_0 = 1$ and, for $j \geq 1$,

$$H_j = \frac{h_j M^j}{h_0} = \frac{h_j \rho^j h_0^j}{h_0} = h_j \rho^j h_0^{j-1} \in \mathbb{Z},$$

using $h_0 \mid M$. Since $\deg h = n$ and $M \neq 0$, $\deg H = n$.

For W , the coefficient of X^j in $\rho W = v(0)H - \tilde{v}$ is $v(0)H_j - v_j M^j$. At $j = 0$ this is $v_0 \cdot 1 - v_0 = 0$, which gives both integrality there and $W(0) = 0$. For $j \geq 1$ the display above exhibits $\rho^j \mid H_j$, and $\rho \mid M^j$, so ρ divides the whole coefficient. $\square$

Remark 1. For the proof of Theorem 1 it is cleaner to treat H and W as *data* subject to the defining identities

$$h_0 H = \tilde{h}, \quad \rho W = v(0)H - \tilde{v}, \quad H(0) = 1, \quad W(0) = 0, \quad (2)$$

rather than as quotients. Nothing after this point uses anything else about them, and no division in $\mathbb{Z}[X]$ ever has to be justified again. Lemma 1 is then exactly the statement that such data exists.

3 The two identities

Both statements below are exact identities in $\mathbb{Z}[X]$. There are no congruences, no roots and no primality anywhere in this section; in particular *irreducibility of h is never used here*. It is needed only to produce the Bézout data in §2 and again in §5.2.

Dilating the Bézout relation $uh + vh' = \rho$ by the ring homomorphism $p \mapsto \tilde{p}$ gives the form in which it will be used,

$$\tilde{u}\tilde{h} + \tilde{v}\tilde{h}' = \rho, \quad \text{equivalently} \quad \tilde{v}\tilde{h}' = \rho - h_0\tilde{u}H, \quad (3)$$

the second form by $\tilde{h} = h_0H$.

Lemma 2 (Derivative). $\rho g' = HA$, where $A := Mh_0\tilde{u} + Mv(0)\tilde{h}' + \rho h_0W' \in \mathbb{Z}[X]$.

Proof. From $g = MX + \tilde{h}W$ and the chain rule (1),

$$g' = M + M\tilde{h}'W + \tilde{h}W'.$$

Multiply by ρ and substitute $\rho W = v(0)H - \tilde{v}$ and $\tilde{h} = h_0H$:

$$\rho g' = \rho M + M\tilde{h}'(v(0)H - \tilde{v}) + h_0H\rho W' = \rho M - M\tilde{h}'\tilde{v} + H(Mv(0)\tilde{h}' + \rho h_0W').$$

By (3), $\rho M - M\tilde{h}'\tilde{v} = \rho M - M(\rho - h_0\tilde{u}H) = Mh_0\tilde{u}H$, and collecting the factor H gives the claim. $\square$

Lemma 3 (Value). $\rho(f \circ g) = H^2D$ for an explicit $D \in \mathbb{Z}[X]$.

Proof. Put $B := h_0HW$, so that $g = MX + B$. By Taylor to second order (Lemma 4) there is $k \in \mathbb{Z}[X]$ with

$$f \circ g = \tilde{f} + \tilde{f}'B + B^2k.$$

Since $f = hf_1$ we have $\tilde{f} = \tilde{h}\tilde{f}_1 = h_0H\tilde{f}_1$, and $B^2 = h_0^2H^2W^2$, so

$$f \circ g = h_0H(\tilde{f}_1 + \tilde{f}'W) + H^2(h_0^2W^2k). \quad (4)$$

It remains to extract one more factor of H from the first bracket. Dilating the product rule $f' = h'f_1 + hf'_1$ and multiplying by $\tilde{v}$,

$$\tilde{f}'\tilde{v} = \tilde{h}'\tilde{v}\tilde{f}_1 + \tilde{h}\tilde{f}'_1\tilde{v} = (\rho - h_0\tilde{u}H)\tilde{f}_1 + h_0H\tilde{f}'_1\tilde{v},$$

using (3) again. Hence $\rho\tilde{f}_1 - \tilde{f}'\tilde{v} = h_0H(\tilde{u}\tilde{f}_1 - \tilde{v}\tilde{f}'_1)$, and so

$$\rho(\tilde{f}_1 + \tilde{f}'W) = \rho\tilde{f}_1 + \tilde{f}'(v(0)H - \tilde{v}) = HC, \quad C := h_0(\tilde{u}\tilde{f}_1 - \tilde{v}\tilde{f}'_1) + v(0)\tilde{f}'.$$

Multiplying (4) by ρ and substituting gives $\rho(f \circ g) = H^2D$ with $D := h_0C + \rho h_0^2W^2k$. $\square$

Lemma 4 (Taylor to second order). *Let R be a commutative ring, $f \in \mathbb{Z}[X]$ and $a, b \in R$. Then there is $k \in R$ with $f(a + b) = f(a) + f'(a)b + b^2k$.*

Proof. Both sides are additive in f and the statement is stable under multiplication by a constant, so it suffices to check $f = X^m$, where the binomial theorem gives $(a + b)^m = a^m + ma^{m-1}b + b^2 \sum_{i \geq 2} \binom{m}{i} a^{m-i} b^{i-2}$. Applied with $R = \mathbb{Z}[X]$, $a = MX$ and $b = B$, this is the form used in Lemma 3. $\square$

4 Descent: removing ρ

Lemmas 2 and 3 give the desired divisibilities only after multiplication by ρ . Removing it is where the normalization $H(0) = 1$ earns its place.

Lemma 5 (Descent). *Let $P, K \in \mathbb{Z}[X]$ and $\rho \in \mathbb{Z}$, $\rho \neq 0$, and suppose $K(0)$ is a unit of $\mathbb{Z}$. If $K \mid \rho P$ in $\mathbb{Z}[X]$ then $K \mid P$ in $\mathbb{Z}[X]$.*

Proof. By strong induction on the number of prime factors of $|\rho|$. If $\rho = \pm 1$ there is nothing to prove. Otherwise choose a prime $p \mid \rho$ and write $\rho = p\rho_1$, and let $\rho P = KD$. Reducing modulo p gives $0 = \overline{K} \overline{D}$ in $\mathbb{F}_p[X]$. Now $K(0) = \pm 1 \not\equiv 0 \pmod{p}$, so $\overline{K} \neq 0$; and $\mathbb{F}_p[X]$ is an integral domain, so $\overline{D} = 0$, that is $D = pD_1$. Cancelling p in the domain $\mathbb{Z}[X]$ leaves $\rho_1 P = KD_1$, and $|\rho_1|$ has one prime factor fewer. $\square$

Corollary 2. $H \mid g'$ and $H^2 \mid f \circ g$, both in $\mathbb{Z}[X]$.

Proof. Apply Lemma 5 to Lemma 2 with $K = H$, whose constant coefficient is 1; and to Lemma 3 with $K = H^2$, whose constant coefficient is likewise 1. $\square$

Remark 2. This is why the construction normalizes $H(0) = 1$ instead of making H monic. Two other routes reach Corollary 2. One: $H(0) = 1$ makes H a unit in the power series ring $\mathbb{Z}[[X]]$, so $(f \circ g)H^{-2}$ lies in $\mathbb{Z}[[X]]$; it also equals $D/\rho \in \mathbb{Q}[X]$; and $\mathbb{Q}[X] \cap \mathbb{Z}[[X]] = \mathbb{Z}[X]$. Two: H is primitive and irreducible, hence prime in $\mathbb{Z}[X]$, and $H \nmid \rho$ by degree. The second route needs §5.2 first, while the induction above and the power series argument need nothing.

5 Nondegeneracy and irreducibility

5.1 Nondegeneracy

Lemma 6. $W \neq 0$, $\deg W \geq 1$, $\deg g = n + \deg W \geq 2$, and $g' \neq 0$.

Proof. Suppose $W = 0$. Then $\tilde{v} = v(0)H$, and comparing with $h_0H = \tilde{h}$ gives $h_0\tilde{v} = v(0)\tilde{h}$; since dilation is injective, $h_0v = v(0)h$. Then $h \mid h_0v$ in $\mathbb{Q}[X]$, so $v = (v(0)/h_0)h$, and the Bézout relation collapses to $\rho = h(u + (v(0)/h_0)h')$, forcing $\deg h = 0$ and contradicting $n \geq 1$. So $W \neq 0$, and $W(0) = 0$ then gives $\deg W \geq 1$.

Since $M \neq 0$ we have $\deg \tilde{h} = n$, so $\deg(\tilde{h}W) = n + \deg W \geq n + 1 \geq 2 > 1 = \deg(MX)$. There is therefore no cancellation in $g = MX + \tilde{h}W$ and $\deg g = n + \deg W$. Finally $\deg g \geq 2 \geq 1$ and characteristic zero give $g' \neq 0$. $\square$

If the Bézout data is chosen with $\deg v < n$, which is automatic for the subresultant construction, then $\deg W \leq n$ and hence

$$n + 1 \leq \deg g \leq 2n.$$

The lower bound is not an artifact: $H \mid g'$ with $\deg H = n$ forces $\deg g' \geq n$ for *any* solution whatsoever, so no construction can do better. Example 2 attains it.

5.2 Irreducibility

Lemma 7. H is irreducible in $\mathbb{Z}[X]$.

Proof. h is primitive and irreducible in $\mathbb{Z}[X]$, hence irreducible in $\mathbb{Q}[X]$ by Gauss's lemma. For $M \neq 0$ the dilation $X \mapsto MX$ is a $\mathbb{Q}$ -algebra automorphism of $\mathbb{Q}[X]$, so $\tilde{h}$ is irreducible in $\mathbb{Q}[X]$; and $H = \tilde{h}/h_0$ differs from it by a unit of $\mathbb{Q}$, so H is irreducible in $\mathbb{Q}[X]$. By Lemma 1, $H(0) = 1$, so H is primitive. Primitive and irreducible over $\mathbb{Q}$ gives irreducible in $\mathbb{Z}[X]$, by Gauss's lemma again. $\square$

This is the only place Gauss's lemma is used, and $\mathbb{Z}$ enters the argument only here and in the prime factorization of Lemma 5. Everything else goes through over any characteristic zero unique factorization domain.

6 The degenerate case

Suppose $X \mid f$. Then

$$g = X^2, \quad H = X$$

settles Theorem 1: $g' = 2X$ is divisible by H ; writing $f = Xf_2$ gives $f \circ g = X^2(f_2 \circ X^2)$, which is divisible by $H^2 = X^2$; and X is irreducible in $\mathbb{Z}[X]$ of degree 1, with $\deg g = 2$.

Note that the case split is on $X \mid f$, not on f being a monomial: this branch covers $f = X(X^2 + 1)$ as well as $f = cX^k$. Equivalently, the split is on whether f has *some* irreducible factor with nonzero constant term.

Proof of Theorem 1. If $X \mid f$, use §6. Otherwise take h , and Bézout data, as in §2, and define M, H, W, g by the table; these lie in $\mathbb{Z}[X]$ by Lemma 1. Then H is irreducible by Lemma 7 with $\deg H = n \geq 1$, the two divisibilities hold by Corollary 2, and $\deg g \geq 2$ by Lemma 6. $\square$

7 Critical values

Only now do roots appear.

Proof of Corollary 1. Let $\alpha \in \overline{\mathbb{Q}}$. If $\alpha = 0$, take $g = X^2$ and $\beta = 0$. Otherwise let h be the primitive integral minimal polynomial of α , so $h(0) \neq 0$, and run §2 with $f = h$, hence $f_1 = 1$. Set

$$\beta := \alpha/M.$$

Then $H(\beta) = h(M\beta)/h_0 = h(\alpha)/h_0 = 0$. Evaluating the Bézout relation at α and using $h(\alpha) = 0$ gives $v(\alpha)h'(\alpha) = \rho$, so

$$W(\beta) = \frac{v(0)H(\beta) - v(\alpha)}{\rho} = -\frac{v(\alpha)}{\rho} = -\frac{1}{h'(\alpha)},$$

which is defined because h is separable. Therefore

$$g(\beta) = M\beta + h(\alpha)W(\beta) = \alpha, \quad g'(\beta) = M + Mh'(\alpha)W(\beta) + h(\alpha)W'(\beta) = M(1 - 1) = 0.$$

So β is a critical point of g with critical value α . The converse was observed at the start of §1. $\square$

Two features of this are worth naming. First, β is the *same* rescaling α/M for every root α of h simultaneously: a single g has all n conjugates as critical values, at the n points α_i/M . Second, the ring at work is $\mathbb{Z}[\beta] = \mathbb{Z}[\alpha/M] \supseteq \mathbb{Z}[\alpha][1/\rho]$. Dividing α by M is what makes the higher Taylor coefficients integral, while $H(0) = 1$ makes β a *unit* in $\mathbb{Z}[\beta]$, and that is what permits trading the constant term of $1/h'(\alpha)$ for higher degree terms.

What does *not* appear is as much to the point: no class group of $\mathbb{Q}(\alpha)$, no unit group $\mathcal{O}_K^\times$, no monogenicity of $\mathcal{O}_K$, and no Thue equation or index form. $\mathbb{Z}[\beta]$ is not an order in $\mathcal{O}_K$ — β is not an algebraic integer — and the units the argument consumes are S -units manufactured by the construction itself.

For a reducible f the statement reduces to the irreducible case for a reason worth stating explicitly, since it is easy to mis-handle: a root α of f is a root of *some* irreducible factor h of f , and the construction run on that h produces a g having α as a critical value. Different factors give different g . It is not the case that one g serves all of a reducible f .

8 Examples

Each line below is a closed identity, checkable by expansion.

Example 1 ($f = qX - p, p \neq 0$). With $u = 0, v = 1, \rho = q, h_0 = -p, M = -pq, H = q^2X + 1$ and $W = qX$,

$$g = -pq^3X^2 - 2pqX, \quad g' = -2pq(q^2X + 1), \quad f \circ g = -p(q^2X + 1)^2.$$

Example 2 ($f = X^2 + 1$). With $u = 2, v = -X, \rho = 2, h_0 = 1, M = 2, H = 4X^2 + 1$ and $W = X$,

$$g = 4X^3 + 3X, \quad g' = 3(4X^2 + 1), \quad f \circ g = (4X^2 + 1)^2(X^2 + 1).$$

Here $\deg g = 3 = n + 1$ attains the lower bound of §5.1. Note also $\rho = 2$ while $\text{Res}(h, h') = 4$: the reduced resultant is the better choice. The unit group $\mathcal{O}_K^\times = \{\pm 1, \pm i\}$ has rank 0, and it is $\mathbb{Z}[i/2] = \mathbb{Z}[i][1/2]$ that supplies the needed unit $1 + i$.

Example 3 ($f = X^3 - 2$). With $u = 3, v = -X, \rho = -6, h_0 = -2, M = 12, H = 1 - 864X^3$ and $W = -2X$,

$$g = -3456X^4 + 16X, \quad g' = 16(1 - 864X^3).$$

Example 4 ($f = X^3 - X - 1$). The construction gives $M = 23$ and a sextic. A hand-tuned quintic also witnesses the *statement*, and is a useful independent check of it:

$$g = -24334X^5 + 39675X^4 - 26450X^3 + 9085X^2 - 1610X + 118, \quad H = 23X^3 - 23X^2 + 8X - 1,$$

with $g' = -230(23X - 7)H$ and $H^2 \mid f \circ g$, the cofactor having degree 9.

Example 5 (Dedekind's cubic). $f = X^3 - X^2 - 2X - 8$, the standard example of a field whose ring of integers is not monogenic: 2 is a common index divisor [Ded78]. With $u = 21X - 125, v = -7X^2 + 44X - 3, \rho = 1006, h_0 = -8, M = -8048$,

$$\begin{aligned} H &= 65158925824X^3 + 8096288X^2 - 2012X + 1, & W &= -194310912X^3 + 426544X^2 + 358X, \\ g &= 101288722414414331904X^6 - 209759614012620800X^5 \\ &\quad - 217370176548864X^4 - 14767629312X^3 + 2350016X^2 - 10912X. \end{aligned}$$

9 Formalization

The development above has been formalized in Lean 4 [MU21] against Mathlib [The20], in about 1,200 lines. The main statement is

```
1 theorem main (f : Z[X]) (hf : 1 <= f.natDegree) :
2   exists g H : Z[X], Irreducible H /\ H \mid derivative g /\ H^2 \mid f.comp g
```

with a strengthened form carrying $2 \leq \deg g$ and $1 \leq \deg H$, and with Corollary 1 proved from it. Every result depends only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

The organization follows the sections above: the dilation and its chain rule (1) first, then `Setup`, a structure carrying the data of §2 with (2) as hypotheses, then Lemmas 2 and 3 as `linear_combination` calls from those hypotheses, then descent, irreducibility and nondegeneracy. Treating H and W as data rather than as quotients, as in Remark 1, is what keeps §3 free of any reasoning about divisibility in $\mathbb{Z}[X]$; the integrality of Lemma 1 is discharged once, in the existence proof for `Setup`.

Three things were genuinely easier on paper than in Lean, and are recorded in case they are useful to someone attempting a similar development. Dilation as a ring homomorphism, and its interaction with `derivative`, is used everywhere and deserves to be isolated first. Lemma 4 is stated in Mathlib for `eval` rather than for `comp`, and has to be transported. And Mathlib has the translation $X \mapsto X + c$ as an algebra equivalence but not the dilation $X \mapsto cX$, which had to be built to prove Lemma 7.

References

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